

WEEKLY REPORT :

Project Title: Self Driving RC Car

Date: 9/29/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

Project Members:

- Ethan Wells (Group Leader)
- Alexis Perez
- Ethan Bishop
- Abigail Duran

Weekly Summary (9/29/2026):

- This week the team finished bench-testing the core hardware on the Raspberry Pi and moved the sign-recognition model onto US traffic signs. Ethan Wells completed both of last week's machine-learning goals: he prepared the LISA Traffic Signs dataset (US signs) for training, writing `prepare_lisa.py` to validate the download, crop each annotated sign, and split the crops into training, validation, and test sets, narrowing the dataset's 47 sign classes to the 12 the car needs (6,097 cropped signs) and keeping every frame of the same physical sign in one split so near-identical images cannot appear in both training and testing; he also tested the German-sign (GTSRB) model on the 12,630-image GTSRB test set, where it reached 95.8% accuracy. Alexis Perez researched and presented the vehicle navigation options the team can use to follow the course, and completed an OpenCV course in preparation for the camera and image-processing pipeline. Ethan Bishop finished testing the Raspberry Pi HAT motor controllers for operational accuracy, running both HATs over I2C on the 40-pin GPIO header, and finished implementing the IMU, the sensor that reports the vehicle's heading and turns, over I2C and verified its measurements and directions; research on 3D-printable chassis files carries over to next week. Abigail Duran connected and communicated with the LCD status display and the camera from the Raspberry Pi Compute Module 5, and integrated both with the

machine-learning model so detected signs and their corresponding commands are shown on the display.

Proposed Plan for Next Week:

- Ethan Wells: Train a CNN on the prepared 12-class LISA dataset (US traffic signs) and test it on the LISA test split.
- Alexis Perez: Research and purchase any components still missing for the selected navigation option, and continue learning OpenCV for the camera pipeline.
- Ethan Bishop: Research 3D-printable STL files for the chassis and research a possible ROS implementation for the vehicle.
- Abigail Duran: Begin planning the PCB layout, including component placement and board dimensions, and begin planning power distribution and signal routing.

Weekly Contributions:

- Ethan Wells:
- 9/26/2026 – Downloaded the LISA Traffic Signs dataset from the Kaggle mirror after the UCSD download host was unavailable, and began the dataset preparation script (2 hours)
- 9/28/2026 – Finished prepare_lisa.py, which validates the dataset, crops each annotated sign, narrows the 47 classes to the 12 the car needs, and splits the 6,097 crops by sign track into training, validation, and test sets; documented the steps in LISA_PREPARATION.md (2 hours)
- 9/29/2026 – Tested the German-sign (GTSRB) model on the 12,630-image GTSRB test set, reaching 95.8% accuracy, and saved the results to gtsrb_test_results.json (3 hours)
- Alexis Perez:
- Researched, finalized, and presented the vehicle navigation options
- Completed an OpenCV course in preparation for the camera and image-processing pipeline
- Ethan Bishop:

- Tested the Raspberry Pi HAT motor controllers for operational accuracy, running both HATs over I2C on the 40-pin GPIO header
- Implemented the IMU on the Raspberry Pi over I2C and verified its measurements and directions
- Abigail Duran:
- Connected and communicated with the LCD and camera using the Raspberry Pi Compute Module 5
- Integrated the LCD and camera with the machine-learning model to display detected signs and their corresponding commands
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Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	7	26
Alexis Perez	7	23
Ethan Bishop	7	22
Abigail Duran	6	23